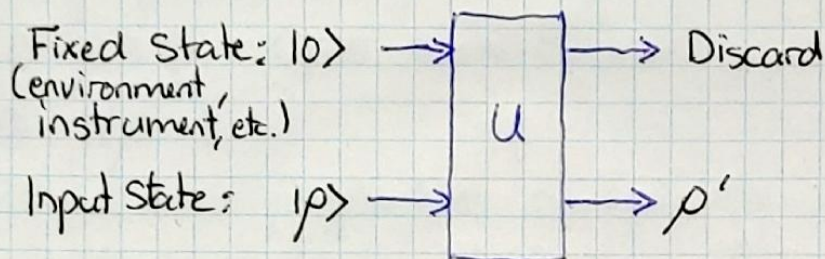


Stinespring/Kraus to CPTP Maps and Choi Matrices

Lec. 16 - Jan 22/26 ⁽¹⁾

We have this picture of a quantum state evolving under a measurement:



- we are only interested in a partial subsystem of a whole.
- this gave rise to two equivalent representation of the evolution, $\rho \rightarrow \rho'$

$$\textcircled{1} \rho \rightarrow \text{tr}_Y V \rho V^\dagger$$

$$\textcircled{2} \rho \rightarrow \sum_k K_k \rho K_k^\dagger$$

where $V^\dagger V = I_X$ (Stinespring)

where $\sum_k K_k^\dagger K_k = I_X$ (Kraus).

NOTATION: - we will ignore internal d.o.f. in the ancilla, $Y \rightarrow$ No Y_{env} .

- we have Kraus operators K_k on system space X .

- the selective measurement map for outcome k :

$$\begin{aligned} \mathcal{E}_k(\rho) &= K_k \rho K_k^\dagger & (\mathcal{E}_k: L(X) \rightarrow L(X)) \\ &\equiv \tilde{\rho}_k & (\text{unnormalized post-meas. state}) \end{aligned}$$

- probability: $p(k) = \text{tr} \tilde{\rho}_k$

- normalized conditional outcome state:

$$\rho_k = \frac{\tilde{\rho}_k}{p(k)}$$

Non-selective (ignore outcome) state update:

- effects operators $E_k = K_k^\dagger K_k$

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$$

- for now only consider a non-selective state update:

$$\rho \rightarrow \rho' = \mathcal{E}(\rho) \quad (\mathcal{E}: \mathcal{L}(X) \rightarrow \mathcal{L}(X))$$

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$$

- a density operator is just a map on X , such that:

① $\rho \geq 0$ (pos. semi-definite).

② $\text{tr} \rho = 1$

→ these two properties completely define a density op.

- so the output state must also have these properties.

So, ① $\mathcal{E}(\rho) \geq 0$

"map is positive"

② $\text{tr} \mathcal{E}(\rho) = 1$

"map is trace preserving"

- we also need: ③ the map $\mathcal{E}(\cdot)$ to be linear in ρ .

Why? → a probabilistic argument.

- the state $\rho = q\rho_1 + (1-q)\rho_2$ can be considered the state where the system is in state ρ_1 with prob. q and ρ_2 with prob. $(1-q)$.

- the output state is:

$$\mathcal{E}(q\rho_1 + (1-q)\rho_2).$$

- this must equal the state $\mathcal{E}(\rho_1)$ with prob. q and ρ_2 with prob. $(1-q)$

$$\Rightarrow \mathcal{E}(q\rho_1 + (1-q)\rho_2) = q\mathcal{E}(\rho_1) + (1-q)\mathcal{E}(\rho_2).$$

- and so $\mathcal{E}(\cdot)$ is linear in ρ in general.

③ ✓

→ easy to check the Kraus expansion satisfies all three req'ts

There is an extra condition, though, that is required: ③

⇒ Completely Positive

- this extra condition arises when we consider adding an extra vector space to X (e.g., an ancilla or reference).

→ this can break the positivity of a map.

• note the $\mathcal{E}$ as a linear operator itself maps linear operators on X to linear operators on X .

$$\text{i.e., } \mathcal{E}: L(X) \longrightarrow L(X)$$

$$\rho \longmapsto \rho' (= \sum_k K_k \rho K_k^\dagger \text{ in Kraus operator form})$$

• you can see this explicitly in the Kraus expansion by taking the $\text{vec}()$ of both sides.

- as an operation on matrices this just stacks the columns into an $n^2 \times 1$ vector

• use the identity for vec and Kronecker products:

$$\text{vec}(AXB) = (B^T \otimes A) \text{vec } X$$

• so, $\text{vec}(K_k \rho K_k^\dagger) = (K_k^* \otimes K_k) \text{vec } \rho$ (term by term)

$$\Rightarrow \text{vec } \rho' = \left(\sum_k (K_k^* \otimes K_k) \right) \text{vec } \rho = [\mathcal{E}] \vec{\rho}$$

$\uparrow$
 $n^2 \times 1$

$\underbrace{\hspace{1cm}}$
a large $n^2 \times n^2$
matrix

$\uparrow$
 $n^2 \times 1$ vector

→ an explicit linear transform in matrix vector form

• issue can arise when a positive mapping $L(X) \rightarrow L(X)$ is not positive on an extended system.

→ we can have: $\Phi: L(X) \rightarrow L(X) > 0$

but $(\text{id}_Y \otimes \Phi): L(Y \otimes X) \rightarrow L(Y \otimes X)$ is not > 0
 $\nwarrow$ do nothing.

Example: transpose operation.

(4)

- let $X = \mathbb{C}^d$

- define the transpose map on operators on X

$$T: L(X) \rightarrow L(X)$$

$$A \mapsto T(A) = A^T$$

- NOTE: ① is linear

② takes pos. operators to positive operator
(eigenvalues don't change)

③ preserves trace "∵" diagonal elements don't change

- BUT: It is not completely positive!

⇒ we need $\text{id}_Y \otimes T$ to be positive for every possible

Y (of some dimension) and

$$(\text{id}_Y \otimes T)(\rho_{YX}) \geq 0 \text{ for every } \rho_{YX} \in L(Y \otimes X).$$

- take $X = \mathbb{C}^2$ and $Y = \mathbb{C}^2$

- let $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \in Y \otimes X$

- let $\rho_{YX} = |\Phi\rangle\langle\Phi|$

→ is semi-pos. definite "∵" its eigenvalues are 0 and 1
(a rank-1 projector).

- now apply the partial transpose operation on ρ_{YX} .

- write out ρ_{YX} in its full form:

$$\rho_{YX} \propto (|00\rangle + |11\rangle)(\langle 00| + \langle 11|)$$

$$= |00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

- the transpose operator acts on the basis operator/matrices, $|iXj\rangle$, as:

$$T(|iXj\rangle) \longrightarrow |jXi\rangle$$

- so, $I \otimes T$ will swap just the indices of the X components

$$T(|\underline{00}X\underline{00}\rangle + |\underline{00}X\underline{11}\rangle + |\underline{11}X\underline{00}\rangle + |\underline{11}X\underline{11}\rangle)$$

$$= |00X00\rangle + |01X10\rangle + |10X01\rangle + |11X11\rangle$$

$$= \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

← eigenvalues of a block diagonal matrix are the eigenvalues of the individual blocks.

- we get two ± 1 eigenvalues from the corners.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightarrow \boxed{\lambda = \pm 1}$$

⇒ we have a negative eigenvalue

→ $\therefore (I \otimes T)$ is not psd in general while T is.

- $\therefore$ the transpose operation is not a quantum channel

→ it cannot be expressed in Kraus form or as a Stinespring dilation! (Will show this)

Defⁿ: Quantum Channel.

⑥

A quantum channel is a linear map,

$$\Phi: L(X) \rightarrow L(X)$$

that is:

① Completely positive (CP):

For every reference system Y ,

$$\rho_{YX} \geq 0 \text{ mean } (\text{id}_Y \otimes \Phi) \rho_{YX} \geq 0$$

② Trace Preserving (TP):

$$\text{tr } \Phi(A) = \text{tr } A \text{ for all } A \in L(X).$$

Th^m: Kraus- CPTP Equivalence (finite-dimⁿ).

Let X be a finite dimⁿ Hilbert space and a map

$\Phi: L(X) \rightarrow L(X)$ be linear. Then the following are equivalent:

① Φ is CPTP (completely positive and trace preserving)

② There exist operators $K_a: X \rightarrow X$ such that for all $\rho \in L(X)$,

$$i) \Phi(\rho) = \sum_a K_a \rho K_a^\dagger$$

$$ii) \sum_a K_a^\dagger K_a = I_X$$

And CP corresponds to the operator-sum form, ⁱ⁾ and TP corresponds to identity sum, ⁱⁱ⁾.

Proof:

Part A: Kraus form $\Rightarrow$ TP, CP.

① Kraus $\Rightarrow$ TP:

$$\text{Let } \Phi(A) = \sum_a K_a A K_a^\dagger.$$

$$\begin{aligned}\text{Then, } \text{tr } \Phi(A) &= \text{tr } \sum_a K_a A K_a^\dagger \\ &= \sum_a \text{tr}(K_a A K_a^\dagger) \\ &= \sum_a \text{tr}(K_a^\dagger K_a A) \\ &= \text{tr} \left[\underbrace{\left(\sum_a K_a^\dagger K_a \right)}_{= I_X} A \right]\end{aligned}$$

$$\therefore \text{tr } \Phi(A) = \text{tr } A \quad \Rightarrow \text{the Kraus form preserves the trace.}$$

② Kraus $\Rightarrow$ CP:

Side derivations first

① If $\rho \geq 0$ (pos. semi-def.) then $M\rho M^\dagger \geq 0$.

(This is just conjugation, or a co-ord. transformation

$\Rightarrow$ does not change intrinsic properties (like positive, trace, det, etc.).

$$\bullet \rho \geq 0 \rightarrow \langle v | \rho | v \rangle \geq 0 \quad \forall v$$

$$\bullet \text{ but } \langle v | M\rho M^\dagger | v \rangle = \langle M^\dagger v | \rho | M^\dagger v \rangle \geq 0$$

$$\therefore M\rho M^\dagger \geq 0$$

② Let R be any reference system and define $\text{id}_R \otimes \Phi$ on $L(R \otimes X)$:

Then:
$$(I_R \otimes \Phi) \rho_{RX} = \sum_a (I_R \otimes K_a) \rho_{RX} (I_R \otimes K_a^\dagger)$$

-NOTE: Once we show this, if $\rho_{RX} \geq 0$ then each term in the sum is ≥ 0 by ① previous $\Rightarrow$ sum is CP.

Why?

(LHS)

(RHS)

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Want to show: $(I_R \otimes \Phi) \rho_{RX} = \sum_a (I_R \otimes K_a) \rho_{RX} (I_R \otimes K_a^\dagger)$

when $\Phi(\rho_X) = \sum_a K_a \rho_X K_a^\dagger : L(X) \rightarrow L(X)$.

- expand ρ_{RX} in blocks wrt a basis $|i\rangle$ in R .

$$\rho_{RX} = \sum_{ij} |i\rangle\langle j| \otimes \rho_{ij}$$

where each ρ_{ij} is an operator on X . So,

$$\text{LHS} = (I_R \otimes \Phi) \rho_{RX} = \sum_{ij} |i\rangle\langle j| \otimes \Phi(\rho_{ij})$$

$$= \sum_{ij} |i\rangle\langle j| \otimes \sum_a K_a \rho_{ij} K_a^\dagger$$

$$= \sum_a \sum_{ij} |i\rangle\langle j| \otimes K_a \rho_{ij} K_a^\dagger$$

RHS = (for each a -term in sum):

$$(I_R \otimes K_a) \rho_{RX} (I_R \otimes K_a^\dagger)$$

$$= (I_R \otimes K_a) \left(\sum_{ij} |i\rangle\langle j| \otimes \rho_{ij} \right) (I_R \otimes K_a^\dagger)$$

$$= \sum_{ij} |i\rangle\langle j| \otimes K_a \rho_{ij} K_a^\dagger$$

• taking the $\sum_a$ (rhs-terms) we have LHS = RHS.

But with $\rho_{RX} \geq 0$ as a density matrix,

$$\sum_a (I_R \otimes K_a) \rho_{RX} (I_R \otimes K_a^\dagger)$$

is exactly in the form $M \rho M^\dagger$, and is also ≥ 0 for all R .

∴ $(I_R \otimes \Phi)$ with Φ in a Kraus form is completely positive. **[QED]**